Virtual University of Pakistan

BIO301

Essentials Of Genetics

Final term Past Papers

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Course Instructor

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Group of VU Biologists

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What are epigenetic disorders? 10 marks
Write function of vntr ? 5 marks
How mutation alter gene function? 10 marks
What are point mutations and described three chromosomal structure changes? 5 marks
Gene frequency in autosomal dominant trait?2 marks
What are epigentics? 2 marks
Write three epigentic disorders?3 marks

STRP function

epigentics..

causes of epigenetics.

VNTR mean and application..

point mutation.
long questionn: chromosomal non disjunction types and causes..

and how mutation change alter function?
Define point mutation. 2 marks
Define epigenetics? 2 marks
Define reverse, forward and neutral mutation. 3 marks
Write any three mitocondrial mutation deasease. 3 marks
Frameshift mutation. 5 marks.
Write any five basic complication in pattern of inheritance. 5 marks:

What is chromosomal abberation?how it is classified? 10 marks
How mutations alters the gene function?10 mrks
Intronic position with respect to exon(2)
Epigenetic definition (2)
Point mutation and it's classifications (5)
Chromosomal aberration with type (5)
How chromosomal aberation change gene (10)
VNTR ? Application (5)
Fameshift mutations: 2marks
Define epigentics:2marks
3 disorders of Auntosomal dominant disorders: 3marks
Forward, reverse, and neutral mutations: 3marks

Point mutations and enlist 3 structural changes in chromosomes:5marks
5 complications in inheritance:5marks
Modification of epigenetica:10marks
Mutation alter genes expression:10marks
1:Write the function of STR?2 mrks
2:define epigenetics?2 marks
3:write the name of the three genetic disorders? 3 marks
4:How epigenetics occur?3 marks
5:what is the point mutation?enlist the structural mutations?5 mrks
6:what is VNTR?writes its five applications? 5 mrks
7:what is chromosomal abbreastion?how it is classified?10 mrks
8:how mutations alters the gene function?10 mrks
Q1: What is the function of Short Tandem Repeats Polymorphism? 2 marks
Q2: Define Epigenetics? 2 marks
Q3: What are the three main causes for epigenetics? 3 marks
Q4: Give three genetic disorders? 3 marks
Q5: What is chromosomal aberration? Write at least three variations caused due to change in chromosomal structure? 5 marks
Q6:Enlist 5 complications caused due to the inheritance pattern? 5 marks
Q7: What is point mutation? How mutation alter gene function? 10 marks
Q8: Explain Chromosomal non disjunction, its types and cause? 10 marks
(For MCQs prepare ppts very well. 30 out of 40 were exactly from there, other 10 were also from ppts but they were asked in a conceptual way).
Non disjunctio of chromosomes?types?causes?10 marks
Mutations ? Hoe can they alter the function of a gene ? 10 marks
Single nucleotides polymorphism ?
5 complexities of inherited gene?
Mcqs were very confusing .. overall ..whole paper was confusing
1.Write the modification of epi genetics ? 10
2.How mutation alter gene function ? 10
SHORT QUESTION
3.Define epigenetics ? 2
4.Write the common mitochondrial diseases ? 2
5.What is VNTR ? write their 5 application. ? 3
6.What is point mutation and how change in chromosome structure? 3
7.Write rules of mutation ? 3
Epigenetic?
Cause of epigenetic?
Linker and causative SNP?
Point mutation?

Chromosomal aberration 10 marks?
Baqi bhool gya....
Mcq zyada tr mutation waly topic se thy
1.Write the modification of epi genetics ? 10
2.How mutation alter gene function ? 10
SHORT QUESTION
3.Define epigenetics ? 2
4.Write the common mitochondrial diseases ? 2
5.What is VNTR ? write their 5 application. ? 3
6.What is point mutation and how change in chromosome structure? 3
7.Write rules of mutation ? 3
Aik question or thaw o yaad nai h…. time 2 hrs
MCQ"S FROM PPTS.
1.epigenetic modification.10 marks
2- how mutation alter gene function.10 marks
3- define epigentics.2marks
4- enlist three common autosomal dominant disorders.3 marks
5- diffrentiate between farword, reverse and neutral mutations. 3 marks
6- what is point mutation. enlist three variation that occur due to change in chromosomal structure. 5 marks
7- five complications in inheritence pattern.5 marks
ESSENTIAL OF GENETICS 25 FEB 2017 4:30
SUBJECTIVE 40 MARKS
1.Write the modification of epi genetics ? 10
2.How mutation alter gene function ? 10
SHORT QUESTION
3.Define epigenetics ? 2
4.Write the common mitochondrial diseases ? 2
5.What is VNTR ? write their 5 application. ? 3
6.What is point mutation and how change in chromosome structure? 3
7.Write rules of mutation ? 3
Aik question or thaw o yaad nai h…. time 2 hrs
MCQ"S FROM PPTS.

Difference b/w chiasmata & cross over?
Meiosis I nd II?
Prophase I?
Stages of prophase I?
Which cell division occurs in somatic cells?

Oogenisis and spermatogenisis?
and mcqs was from ppt slides totally

define karyotype idiotype,,linkage inheritance , rule organizes my international committe ,
3 genetic disorders , single gene and multifactorial disorders , vntr aflp ,,inheretance ,dominant allele , frequency of dominant alllele, ,
5 basics which cause difficulty in inheretance pattern and all the mcqs were from starting topics
301 Paper 4-03-16

*Diff b/w forward and reverse genetics

*Reason of drawing pedigree
*what is frequency range of autosomal dominant disorder?
*Diff b/w RAPD and AFLP
*Location of intronic sequence with respect to exons
*Rule of mutation nomenclature
*3 mitochondrial diseases
*3 autosomal dominant disorder
*Types of genetic disorders
LONG OF 5 MARKS
*Five complications in inheritance pattern
*what is VNTR? Write five applications of VNTR.
50% objective from first 20 and 50% from 100 to 125

1 Comment

301 Paper 3-03-16
*Difference between linked and causative SNP
*Diff b/w single gene and multifactorial gene
*Diff b/w forward and reverse genetics
*Function of STRP
*Reason of drawing pedigree
*Diff b/w centromere and telomere
*what is frequency range of autosomal dominant disorder?
*Diff b/w RAPD and AFLP
*Diff b/w meiosis I and II
*Location of intronic sequence with respect to exons
*Rule of mutation nomenclature
*3 mitochondrial diseases
*3 autosomal dominant disorder
*Types of genetic disorders
LONG OF 5 MARKS
*Five complications in inheritance pattern
*what is VNTR? Write five applications of VNTR.
types of poymorphisims.,VNTR function ,locus,mitochondrial disorder,autosomal dominant recesseive disorder
Some d!ffr ques of b!03o1

Def!ne nd expla!n m!tot!c appratus?
Why h!gh gene dens!ty !n parokaryouts?

ANS of BIO301 Paper

types of poymorphisims.

Types of Polymorphic DNA Sequences

- RFLP: Restriction Fragment Length Polymorphism
- VNTR: Variable Number Tandem Repeats (8 to > 50 base pairs)

Types of Polymorphic DNA Sequences

- STR: Short Tandem Repeats (1– 8 base pairs).
- SNP: Single Nucleotide Polymorphisms
- AFLP: Amplified Fragment Length Polymorphism,

,locus

locus (plural loci), in genetics, is the specific location or position of agene, DNA sequence, on a chromosome. Each chromosome carries many genes; humans' estimated 'haploid' protein coding genes are 20,000-25,000, on the 23 different chromosomes. A variant of the similar DNA sequence located at a given locus is called an allele. The ordered list of loci known for a particular genome is called a gene map. Gene mapping is the process of determining the locus for a particular biological trait.

,mitochondrial disorder

Common mitochondrial diseases

- Myoclonic epilepsy
- Mitochondrial recessive ataxia
- Leber hereditary optic neuropathy
- Sensory ataxia neuropathy

,autosomal dominant disorder

Autosomal Dominant diseases

 Marfan syndrome

 Huntington's disease

 Retinoblastoma

 Waardenburg syndrome

 Myotonic dystrophy

 Polycystic kidney disease

 Achondroplasia

 Polydactyly

 Heredity hearing loss

recesseive disorder

Common recessive disorders

 Bloom syndrome

 Carpenter syndrome

 Cystic fibrosis

 Thalassemia

 Many forms of mental retardation

 Gaucher's disease

 Glycogen storage diseases

 Rotor syndrome

 Many of eye diseases

Classification of genetic disorders

 Multifactorial disorders

 Single gene disorders

 Chromosomal disorders

 Mitochondrial disorders

 Somatic mutations (cancers)

Karyotype

A karyotype is the number and appearance of chromosomes in the nucleus of a eukaryotic cell. The term is also used for the complete set of chromosomes in a species, or an individual organism

idiotype

He defined idiotype as the set of epitopes on the V region of an antibody molecule, where epitope means an antigenic determinant. He also defined the "paratope" to be that part of an antibody variable region that binds to an antigen.

Single gene can give codominance, incomplete dominance, overdominance

Genetic maps have been used successfully to find the single gene responsible inherited disorders, like cystic fibrosis and muscular dystrophy.

Multifactorial diseases

 Asthma

 Cancers

 Ciliopathies

 Cleft palate

 Diabetes

 Heart diseases

 hypertension

 inflammatory bowel disease

 intellectual disability

 mood disorder

 Obesity

 infertility

Frequency of genetic disorders

Frequency of autosomal dominant disorders

Frequency of autosomal dominant disorders varies between 3 – 9.5 per 1000 individuals

Frequency of autosomal recessive disorders

Frequency of autosomal recessive disorders is between 2-2.5 per 1000 individuals.

Frequency of x-linked disorders

Frequency of X-linked disorders is between 0.5 – 2 per 1000 individuals.

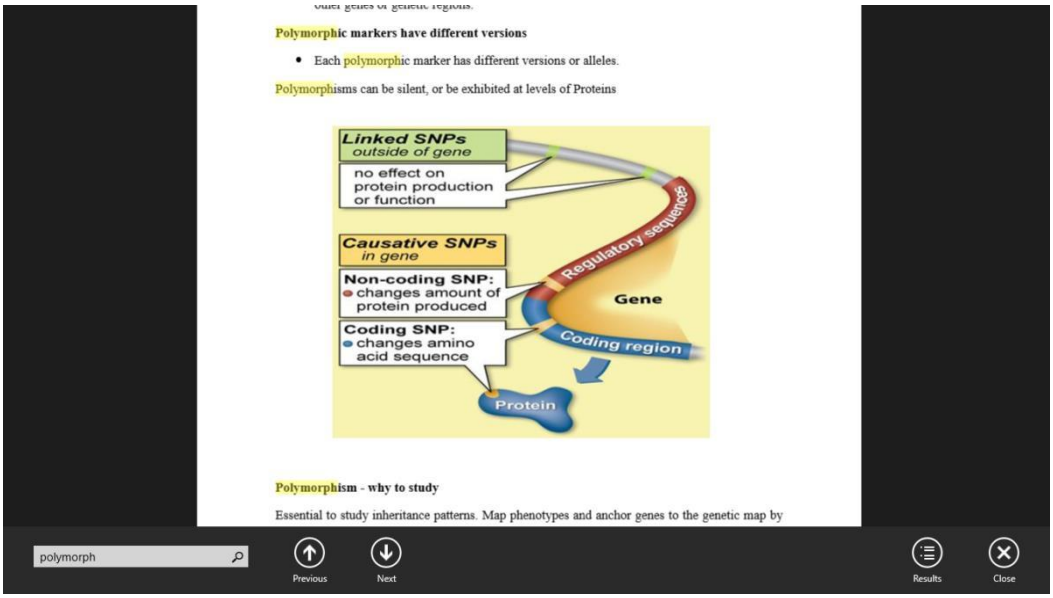
Frequency of chromosomal disorders

Frequency of chromosomal disorders is between 6-9 per 1000 individuals.

Frequency of multifactorial disorders

Frequency of multifactorial disorders is between 20 – 50 per 1000 individuals.

Most common multifactorial disorders are cancers, diabetes, heart diseases.



What is RFLP: RFLP is a difference in homologous DNA sequences that can be detected as fragments of different lengths after digestion of the DNA samples.

Most RFLP markers are heterozygous and highly locus-specific.

RFLP probe is a labeled DNA sequence that hybridizes with one or more fragments of the digested DNA sample after they were separated by gel electrophoresis.

VNTR

 VNTR are short nucleotide sequences organized as a tandem repeats on genomes.

 Found on many chromosomes

 Show variations in length between individuals.

Size of VNTR

 The size of repeat is about 7 to few tens of base pairs long.

Each variant acts as inherited allele

 Each variant acts as an inherited allele, allowing them to be used for personal or parental identification

Importance of VNTR

VNTRs are an important source of RFLP genetic markers used in linkage analysis (mapping) of genomes. When removed from surrounding DNA by the PCR or RFLP methods, and their size determined by gel electrophoresis or Southern blotting. As a result, they produce a pattern of bands unique to each individual.

VNTR function

Applications of VNTR

 Microbiology

 DNA fingerprinting

 Genetic diversity

 Forensics

 Mapping of genomes

 Breeding patterns of wild or domesticated animals

Mutation nomenclature

 All variants should be described at the most basic level, i.e. DNA level.

 Descriptions should always be in relation to a reference sequence, either a genomic or a coding DNA sequence.

 Although theoretically a genomic reference sequence seems best, in practice a coding DNA reference sequence is preferred.

  Good for promoters, alternative splicing etc.

 When the entire genomic sequence is not known, a cDNA reference sequence should be used.

 To avoid confusion in the description of a variant it should be preceded by a letter indicating the type of reference sequence used.

  Several different reference sequences can be used.

 "c." for a coding DNA sequence (like c.76A>T)

 "g." for a genomic sequence (like g.476A>T)

 "m." for a mitochondrial sequence (like m.8993T>C,)

 "r." for an RNA sequence (like r.76a>u)

 "p." for a protein sequence (like p.Lys76Asn)

 Describing genes / proteins, only official HGNC (HUGO gene nomenclature committee) gene symbols should be used (www.genenames.org)

 DNA reference sequence used should preferably be from the RefSeq database, listing both database accession and version number (like NM_004006.2)

Type of Chromosomes based on Centromere positioN Chromosomes may differ in the position of the centromere.

Centromere is near the middle, metacentric

Centromere is toward one end, acrocentric or sub metacentric

Centromere is very near to end, telocentric

pedigree

A complete, detailed family history is called a pedigree. Pedigrees are used to determine the pattern of inheritance of a genetic disease within a family.

Generations of a pedigree as roman numerals

Generations in a pedigree diagram are numbered, by using Roman numerals, starting with the parental generation, at the top of the diagram as generation I.

Members of each generation are numbers from left to right

For convenience, the members of each generation are numbered across the line, from left to right, using normal numerals

Difference B/W Centromere and Telomere?

A centromere is a cellular structure involved in meiosis that creates spindle fibers and allows chromosomes to separate in cells. Telomeres are single-stranded DNA tails that form at the end of replicated DNA strands because they cannot be replicated. They are nonsense DNA, so nothing is lost in not replicating them. They protect exons--expressed DNA segments, and work in the cell aging process.

How mutation alter gene function? 10 marks
What are point mutations and described three chromosomal structure changes? 5 marks..?STRP function?
Define point mutation. 2 marks
Define epigenetics? 2 marks
Define reverse, forward and neutral mutation. 3 marksFrameshift mutation?
Forward, reverse, and neutral mutations?3 genetics disorder

Long questions
Complications to inheritance patterns
Mutation alter gene function
Point mutation and variation structural changes
Non disjunction chromosomes types and causes
Short
Define epigenetics and causes
linked and causative SNP
Frequency of autosomal dominant disorder
frameshift
Mcqs mostly mutation k topic me sy thy

describe complications to inheritance pattern?
How Mutation alter gene function?
what is Point mutation and how variation in structural changes accur?
Non disjunction ? Types of chromosomes ?and causes?
#Short:

Define epigenetics and causes?
what linked and causative SNP?
what is Frequency of autosomal dominant disorder?
define frameshift?

Splicing pathway... CRISTR STANDS FOR??? termination in short... Rna processing .. rna has short life span so how we can analyzed through what process fr elongation short

301 genetics,,,,,, 11.00 am
enlist 3 genetics disoders.point mutatin,chromosoml aberration and cllsfcation.,

how mutation chnge gene function.

definition of forwrd ,reversd and neutral muation,complications of inheritance pattern,intronic location,epigenetic define,
difference b/w single gene disorders and multifactorial disorders. 2m
what is the location of intronic sequences wd respect to exons? 2m
define epigenetics. 2m
diffrentiate b/w RAPD and AFLP. 3m
differentiate b/w forward, reverse and neutral mutation. 3m
wt r basic 5 complications in inheritance pattern. 5m
wt point mutation? enlist at least 3 variations that occur due to change in the structure of chromosome? 5m
write a note on epigenetic modification. 10m
how mutations alter gene function ? 10m

ByBio (301) today paper
How mutation alter gene function? 10 marks
What are point mutations and described three chromosomal structure changes? 5 marks..?STRP function?
Bio 301. 3 march, 2017
1: Define point mutation. 2 marks
2: Define epigenetics? 2 marks
3: Define reverse, forward and neutral mutation. 3 marksFrameshift mutation?
Forward, reverse, and neutral mutations?3 genetics disorder?
Almost all q past paper se thee?

bio 301 final term 11AM
12 aug
36 MCQs
37.....difference b/w single gene disorders and multifactorial disorders. 2m
38.... what is the location of intronic sequences wd respect to exons? 2m
39....define epigenetics. 2m

40.... diffrentiate b/w RAPD and AFLP. 3m
41...differentiate b/w forward, reverse and neutral mutation. 3m
42....wt r basic 5 complications in inheritance pattern. 5m
43.... wt point mutation? enlist at least 3 variations that occur due to change in the structure of chromosome? 5m
44... write a note on epigenetic modification. 10m
45..... how mutations alter gene function ? 10m

- My today paper bio 301 at 11am
  Long questions
  1. Complications to inheritance patterns
  2. Mutation alter gene function
  3. Point mutation and variation structural changes
  4. Non disjunction chromosomes types and causes
  Short
  Define epigenetics and causes
  3.linked and causative SNP
  4. Frequency of autosomal dominant disorder
  5.frameshift
  Mcqs mostly mutation
- topic me sy thy….
- Quiz Bio 301
  Q: 1
  Mutation which type a mutant phenotype back into the wild type.
  Ans: Reverse mutation
  Q: 2
  A genetic disease or disorder appears as a result of mutations in an individual's _____.
  Ans: DNA
  Q: 3
  Hunter's disease is an example of_____.
  Ans: X-linked recessive disorder
  Q: 4
  A condition in which some people with an appropriate genotype fail to express the phenotype is known as the _____.
  Ans: incomplete penetrance
  Q: 5
  Which one of the following is not influenced by the silent mutation?
  Ans: Phenotype
  Q: 6
  Organism that exhibit suppressor mutation is/ are_____

Ans: Double mutant

Q: 7

Mutation that produce the change in the amino acid in protein product (histidine in for arginine)

Ans: Missense mutation

Q: 8

Base analog have the similar structure with_____

Ans: standard bases of the DNA

Q: 9

Mutation which convert a DNA sequence into the stop codon.

Ans: Non sense mutation

Q: 10

The term Ashkenazi refers to the specific people of following population.

Ans: Jew population

- Bio 301paper
-
- Bio 301 ka zaid past ..ma se tha...vNTR or us ke application?2)epigenetics modification?genetics disorder?SNTR function,? mutation alter gene function? Farwad reverse and natural mutation? freamshift mutation define?point mution or sath ma change chromosome structure pouch tha,?epigenetics define?
- subbject: bio 301

Timing: 11am, 12/08/2017

#final term #paper.

#long

1) describe complications to inheritance pattern?

2) . How Mutation alter gene function?

3) what is Point mutation and how variation in structural changes accur?

4) Non disjunction ? Types of chromosomes ?and causes?

#Short:

1) Define epigenetics and causes?

2) what linked and causative SNP?

3) what is Frequency of autosomal dominant disorder?

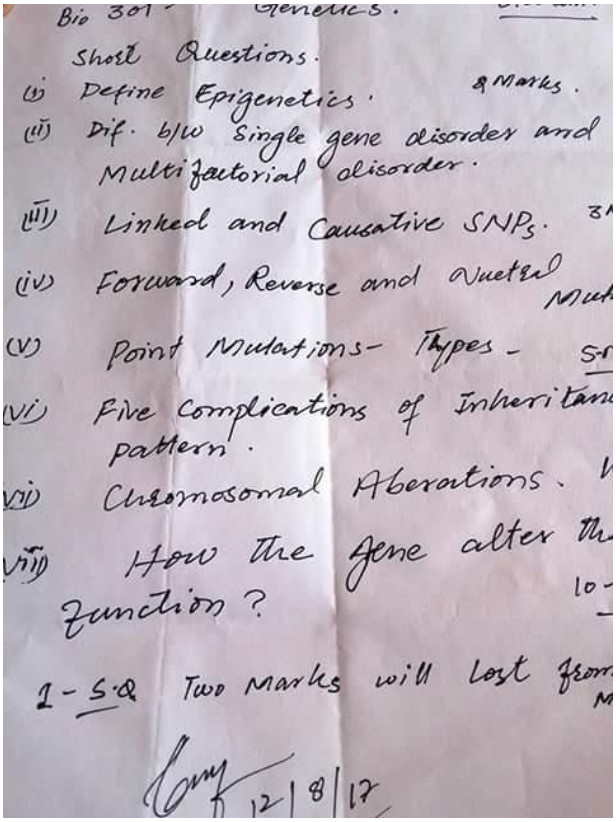
4) define frameshift?

- 301 Genetics ...timming 11.00 (final term)Enlist 3 genetic disorders.Point mutation.structural muation of chromosomes.Chromosomal abetration and classfy them.How mutation change the gene function.5 basic complications of inheritance pattern.Define :Forward.reversed and neutral muation.

301 genetics,,,,,, 11.00 am
enlist 3 genetics disoders.point mutatin,chromosoml aberration and cllsfcation.,how mutation chnge gene function. definition of forwrd ,reversd and neutral muation,complications of inheritance pattern,intronic location,epigenetic define,?

Bio301 final term paper 12-8-17 at 8:00 am
What is Choromosomal non-disjunction types, causes. 10
Mutations alter gene function 10
VNTR and its applications 5
Point mutation 5
Causes of epigenetics 3
Write the name of 3 autosomal resesive disorders 3
Dif. b/w multifactorial and singal gene mutation 2
What is framshift mutation 2

Bio 301 paper…..


Bio 301 - Genetics.

Short Questions.

(i) Define Epigenetics. 2 Marks.

(ii) Dif. b/w Single gene disorder and Multifactorial disorder.

(iii) Linked and Causative SNPs.

(iv) Forward, Reverse and Nuetral Muta

(v) Point Mutations- Types- Sr

(vi) Five Complications of Inheritan pattern.

(vii) Chromosomal Aberations.

(viii) How the gene alter th function ?

1-S.Q Two Marks will Lost from

12/8/17

12 aug

36 MCQs

37.....difference b/w single gene disorders and multifactorial disorders. 2m

38.... what is the location of intronic sequences wd respect to exons? 2m

39....define epigenetics. 2m

40.... diffrentiate b/w RAPD and AFLP. 3m

41...differentiate b/w forward, reverse and neutral mutation. 3m

42....wt r basic 5 complications in inheritance pattern. 5m

43.... wt point mutation? enlist at least 3 variations that occur due to change in the structure of chromosome? 5m

44... write a note on epigenetic modification. 10m

45..... how mutations alter gene function ? 10m

bio 301 final term 11AM

12 aug

36 MCQs

37.....difference b/w single gene disorders and multifactorial disorders. 2m

38.... what is the location of intronic sequences wd respect to exons? 2m

39....define epigenetics. 2m

40.... diffrentiate b/w RAPD and AFLP. 3m

41...differentiate b/w forward, reverse and neutral mutation. 3m

42....wt r basic 5 complications in inheritance pattern. 5m
43.... wt point mutation? enlist at least 3 variations that occur due to change in the structure of chromosome? 5m
44... write a note on epigenetic modification. 10m
45..... how mutations alter gene function ? 10m

BIO301 TODAY'S PAPER
Function of short tendem repeat polymorphism (2)
Is there are any rules for mutation nomenclature? (2)
Define epigenetics (2)
Write three causes of epigenesis (3)
Enlist minimum three mitochondrial diseases (3)
Write basic complications in inheritance pattern (5)
Define point mutation. Enlist three variations caused by changes in chromosome. (5)
Write a note on epigenetic mutations (10)
How mutations alter protein function (10)

BIO301 TODAY'S PAPER
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Define point mutation. Enlist three variations caused by changes in chromosome. (5)
Write a note on epigenetic mutations (10)
How mutations alter protein function (10)

Bio 301 ...16/8/2017.....8:00 AM
1.What is Frameshift mutation? 2
2.Define epigenetics? 2
3.Diff. B/w single gene disorder and multifactorial disorders? 2
4.Enlist three mitochondrial disease? 3
5.Name of cuase of epigenetics? 3
6.what are the basic complications in inheritance pattren? 5
7.Define point mutation? Enlist the variations occur through the chnage in dna structure? 5
8.Explain chromosomal aberrationss? 10
9.How mutation alter the gene functions? 10

16 August 301 paper 8 am
Q1.Define eoigenetics.2m
Q2.Write frequency of dominenet inheritance,2 marks

Q3.Define VNTR and write its applications.5m
Q4.Define point mutation enlistat least three variationsthat occurs due to change in the chromosome structure.5

How mutations change the gene function.10m

How mutation alters gene function. 10
What is chromosomal abberations. Its types. 10
Define VNTR. And its applications 5
Name 3 mitochondrial diseases. 3
Three cause of epigenetics. 3
Define point mutation. 2
Define frame shift mutation. 2

Bio 301 final term paper::::

What is Choromosomal non-disjunction types, causes. 10
Mutations alter gene function 10
VNTR and its applications 5
Point mutation 5
Causes of epigenetics 3
Write the name of 3 autosomal resesive disorders 3
Dif. b/w multifactorial and singal gene mutation 2
What is framshift mutation 2

Bio 301 paper

Mery paper mein.
Alterations of gene functions
Epigenetics modification
Cause of epigenetics
Epigenetic diseases
Autosomal recessive disease
Complication in inheritance pattern?

Bio 301 mcqs at 5pm

1 : Variation is induced by genetic and environment factor?
(a) Height
2 : Mendle work was based on ?
(a) Single gene (b) Two law (c) Both (d) Three law

3 : Color of hydrangea plant change on?
(a) Texture (b) PH (c) composition (d) soil
4 : Bacterial chromosomes is present in ?
(a) Mitochondria (b) Nucleus (c) Nucleoid (d) Nucleolus
5 : Bacterial chromosomes is made of :
(a) DNA (b) Histone (c) Both (d) RNA
6 : Chromosomal is unable to work due to mutation in ?
(a) CDFI (b) CDFII (c) CDFIII (d) CDFIV
7 : Okazaki fragment are farmed ?
(a) Discontinous (b) RNA replication (c) Continuous (d) Conservative
8 : If the gene closely related then ?
(a) Recombination frequency will be constant (b) Recombination frequency will be low
(b) Recombination frequency will be high (d) Recombination frequency will be interior
9 : One map is parallal is produced recombination frequency ?
(a) 1% (b) 2% (c) 3% (d) 4%
10 : location of genes on chromosomes ?
(a) Synapse (b) Chiasmata (c) Locus (d) Tetrad
11 : About 25000 SNA in human are ?
(a) Non – Synapse
12 : Determination of pattern inheritance ?
(a) Pedigree
13 : Types of single nucleoide polymorphism that change the sequence of amino acid ?
(a) Non – coding
14 : Expression of phenotype :
(a) Incomplete inheritance
15 : Small case for alphabets used for ?
(a) RNA level
16 : Frequency of autosomal recessive ?
(a) 2- 2.5
17 : Medelian disorder is :
(a) Single gene
18 : Addition mutation is addition of ?
(a) Nucliotide
19 : Mutation that transfer next generation ?
(a) Somatic mutation (b) Germ line mutation (c) Farword (d) Reverse
20 : Which is one of chromosomal duplication ?
(a) Retro – transposition
21 : Turner syndrome is an example of ?
(a) Monosomy
22 : Suppressor mutation is ?

(a) Double
23 : Gene switch on ?
(a) Deacylyated Not conferm
24 : Two genes close together , frequency of crossing over would be ?
(a) Low

25 : Large arm of chromosomes is ?
(a) Q arm
26 : Colchicine to promoter ?
(a) Polyploidy
27 : Histone deacytylation inhibitor ?
(a) Varinostat (b) DDT (c) Colchicine (d) Tetracycline
28 :Polygenic inheritance is controlled by ?
(a) Multiple genes
29 : Fur color in Labrador is controlled by ?
(a) 2 genes
30 : cancer , infertility are result of ?
(a) Translocation
31 : Which is following is not chromosomal aberration ?
(a) Translation
32 : which is result of frame shift ?
(a) Deletion
33 : Which is not true recessive trait ?
(a) Females are more affected
34 : Molecular biology , forensic , genes mapping are application of ?
(a) VNTR
35 :Mutation can occur regulatory region that effect ?
(a) Transcription , translation or splicing of the protein
36 : ‘’(Underscore) indicate ?
(a) Arange of affected residue

Bio301 18august 11am final term
Total que. 45 (Mcq’s 36x1 + 3x2 + 2x3 + 2x5 + 2x10)
Mcq’s
1.Incomplete dominance, 2.okazaki fragement are formed in,
3.Phase at which spindle fibre break,a. telophase b.metaphase c. yad ni d. yad ni
4.Primary oocyte produces by,
5.Equational division is,a.metaphase b. tekophase c.meiosis I d.meiosis II
6.Sex determination system in fishes a.x-o b.w-z c.x-y d. none
7. Frequency of autosome recessive per 1000 individual, a.0.5-2 b.2-2.5 c.4-9.5 d.20-50
8.Small case-alphabts are use 4 dscrbng mutation at, a. DNA levl b. RNA levl c.protein lvl d.

crbohydrt lvl
9.Addition mutation is the addition of, 10. Rurner syndrome is an example of,
11.A two genes are more close, frequency of crosing over will be, a. high b. low c. high and low d. none
12.Colchicine produces, 13. Polygenetic inherited controlled by,
Short que ….. What is the function of drawing pedigree? (2m) write the location of the intronic sequence with respect to axons? (2m) Define Epigenetics? (2m) differentiate b/w linked and causative nucleotide polymorphism. (3m) Define forward, reverse and neutral mutation. (3

) What is VNTR?
Long que….. Give at least it's 5 application. (5m) Write a note on epigenetics modification. (10)
Write a note on mutation altar the gene function. (10m)
wish all of u best of luck□□□□□

Bio 301

My today paper bio 301 at 11am
Long questions
1. Complications to inheritance patterns
2. Mutation alter gene function
3. Point mutation and variation structural changes
4. Non disjunction chromosomes types and causes
Short
Define epigenetics and causes
3.linked and causative SNP
4. Frequency of autosomal dominant disorder
5.frameshift
Mcqs mostly mutation k topic me sy thy

Bio 301

Today mine bio301 paper at 8 am
total 47 questions 37 mcqs
define epigenetics 2 structural abnormalities 2 reverse forward and neytral mutations 3 non disjunction types abnormalities and causes 5 point mutations 10 frameshift mutation 2 mark
how mutations can alter gene function 10autosomal recessive inheritance 5 explain complication

in inheritance pattern any 5 3 mark
All the best for exam

Bio 301

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All the best for exam

Bio (301) today paper
How mutation alter gene function? 10 marks
What are point mutations and described three chromosomal structure changes? 5 marks..?STRP function?
Bio 301. 3 march, 2017
1: Define point mutation. 2 marks
2: Define epigenetics? 2 marks
3: Define reverse, forward and neutral mutation. 3 marksFrameshift mutation?
Forward, reverse, and neutral mutations?3 genetics disorder?
Almost all q past paper se thee?
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